

Math 491

Homotopy Classification of Vector Bundles

Yusuf Emre Baran

Supervisor: Özgün Ünlü

Department of Mathematics

Bilkent University

Fall 2025

Outline

- Introduction and Examples
- Classification of Vector Bundles
- Sections and Triviality
- Characteristic Classes

Introduction

Definition

A k -dimensional vector bundle ξ is a triple (E, p, B) , where $p : E \rightarrow B$ is a continuous map with a real k -dimensional vector space structure on $p^{-1}(b)$ for each $b \in B$, such that the following condition is satisfied: Each point $b \in B$ has a neighborhood U for which there exists a homeomorphism $h : p^{-1}(U) \rightarrow U \times \mathbb{R}^k$ taking $p^{-1}(b')$ to $\{b'\} \times \mathbb{R}^k$ by a vector space isomorphism for all $b' \in U$.

Example

Let B be a space and $E = B \times \mathbb{R}^k$. Then, $p : E \rightarrow B$ given by

$$p(b, v) = b$$

is a k -dimensional vector bundle, called the trivial bundle over B .

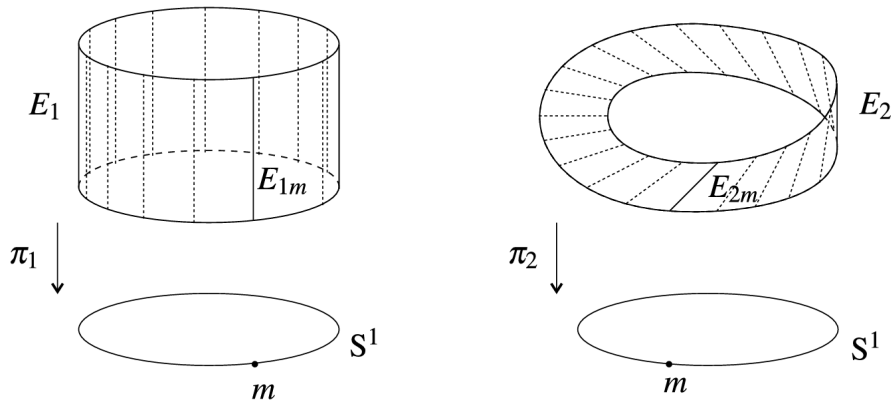
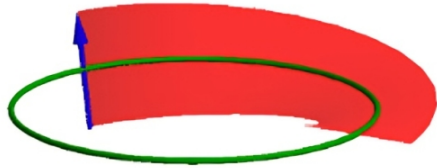
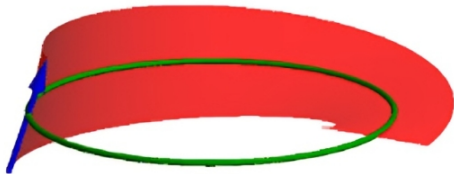
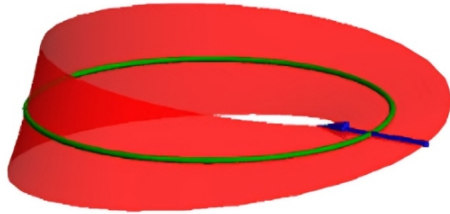


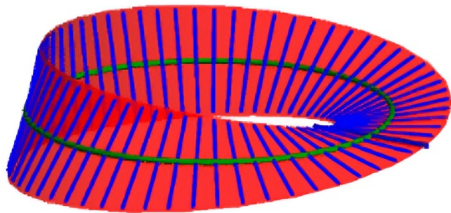
Figure 1: The cylinder and the Möbius strip as a vector bundle over S^1 [3]

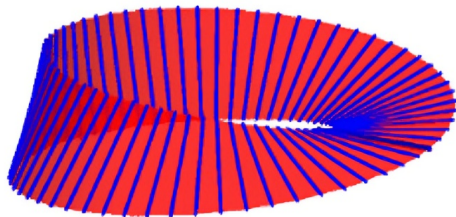












Isomorphisms of Vector Bundles

How many different k -dimensional vector bundles can exist over a given space B ?

Definition

An isomorphism between vector bundles $p_1 : E_1 \rightarrow B$ and $p_2 : E_2 \rightarrow B$ is a homeomorphism $h : E_1 \rightarrow E_2$ such that the diagram

$$\begin{array}{ccc} E_1 & \xrightarrow{h} & E_2 \\ & \searrow p_1 & \swarrow p_2 \\ & B & \end{array}$$

commutes and h takes $p_1^{-1}(b)$ to $p_2^{-1}(b)$ by a linear isomorphism.

Example: The Tangent Bundle of the Circle

Let $\xi_1 = (TS^1, p_1, S^1)$ and $\xi_2 = (S^1 \times \mathbb{R}, p_2, S^1)$, where $p_1(b, v) = b \in S^1$. A tangent vector v to S^1 at $b = (\cos \theta, \sin \theta)$ has the form $v = t(-\sin \theta, \cos \theta)$ for some $t \in \mathbb{R}$. Then, $h : S^1 \times \mathbb{R} \rightarrow TS^1$ given by $h(b, t) = (b, v)$ is an isomorphism.

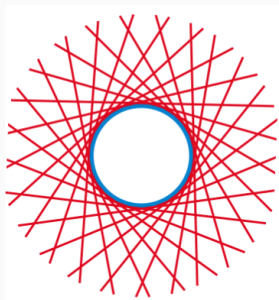


Figure 3: $TS^1 \cong S^1 \times \mathbb{R}$ [4]

Grassmannians

For nonnegative integers $k \leq n$, the Grassmann manifold $G_k(\mathbb{R}^n)$ is the set of all k -dimensional subspaces of $\mathbb{R}^n$.

The Infinite Grassmannian

The natural inclusions $\mathbb{R}^n \subseteq \mathbb{R}^{n+1} \subseteq \dots$ induce inclusions $G_k(\mathbb{R}^n) \subseteq G_k(\mathbb{R}^{n+1}) \subseteq \dots$, and we set

$$G_k(\mathbb{R}^\infty) = \bigcup_{n \geq k} G_k(\mathbb{R}^n).$$

We also define

$$E_k(\mathbb{R}^\infty) = \{(V, x) \in G_k(\mathbb{R}^\infty) \times \mathbb{R}^\infty \mid x \in V\}.$$

Proposition

The projection map $p : E_k(\mathbb{R}^n) \rightarrow G_k(\mathbb{R}^n)$ given by $p(V, x) = V$ is a vector bundle denoted by γ_k^n .

Proof

For $V \in G_k(\mathbb{R}^n)$, let $\Pi_V : \mathbb{R}^k \rightarrow V$ be the orthogonal projection and set

$$U_V = \{V' \in G_k(\mathbb{R}^n) \mid V^\perp \cap V' = 0\} = \{V' \in G_k(\mathbb{R}^n) \mid \dim(\Pi_V(V')) = k\}.$$

Letting A be the matrix of Π_V , we see that the $k \times k$ matrix with columns $Av_1, \dots, Av_k$, where $v_1, \dots, v_k$ is a k -frame, has nonzero determinant. Because the map $(v_1, \dots, v_k) \mapsto (Av_1, \dots, Av_k) \mapsto \det(Av_1, \dots, Av_k)$ is continuous, U_V is open in $G_k(\mathbb{R}^n)$. Then, the map $h : p^{-1}(U_V) \rightarrow U_V \times V \cong U_V \times \mathbb{R}^n$ given by $h(V', x) = (V', \pi_V(x))$ is a local trivialization.

Pullback Bundles

Proposition

Let $\xi = (E, p, B)$ be a vector bundle. For a map $f : A \rightarrow B$, there exists a vector bundle $p' : E' \rightarrow A$ and a map $f' : E' \rightarrow E$ taking $(p')^{-1}(a)$ isomorphically onto $p^{-1}(f(a))$ for each $a \in A$. Moreover, E' is unique up to isomorphism.

We usually write $f^*(E)$ or $f^*(\xi)$ instead of E' and call it the pullback of E by f . More explicitly, E' is given by

$$E' = \{(a, e) \in A \times E \mid f(a) = p(e)\},$$

and $p' : E' \rightarrow A$ is given by $p'(a, v) = a$.

Classification of Vector Bundles

Notation

$Vect_k(B)$ = the set of isomorphism classes of k -dimensional vector bundles over B .

$[X, Y]$ = the set of homotopy classes of maps from X into Y .

The Main Theorem

For a paracompact space B , the map

$$\Phi : [B, G_k(\mathbb{R}^\infty)] \rightarrow Vect_k(B)$$

given by $\Phi([f]) = [f^*(\gamma_k)]$ is a bijection.

In other words, vector bundles over B are classified by homotopy classes of maps into $G_k(\mathbb{R}^\infty)$.

Proposition

Let $f, g : A \rightarrow B$ be homotopic maps, where A is paracompact, and let $p : E \rightarrow B$ be a vector bundle. Then, $f^*(E)$ and $g^*(E)$ are isomorphic.

This follows from the following fact:

Let $r : B \times I \rightarrow B \times I$ be given by $r(b, t) = (b, 1)$, where B is paracompact. Suppose $\xi = (E, p, B \times I)$ is a vector bundle. There exists a map $u : E \rightarrow E$ such that $(u, r) : \xi|_{B \times \{0\}} \rightarrow \xi|_{B \times \{1\}}$ is an isomorphism.

Proof of proposition

Let $f \simeq g$ via $h : A \times I \rightarrow B$. Note that

$$f^*(E) = \{(a, e) \in A \times E \mid f(a) = p(e)\}.$$

Similarly,

$$(h^*(E))|_{A \times \{0\}} = \{((a, 0), e) \in (A \times \{0\}) \times E \mid h(a, 0) = f(a) = p(e)\}.$$

Define $\phi : f^*(E) \rightarrow (h^*(E))|_{B \times \{0\}}$ by $\phi(b, e) = ((b, 0), e)$. It can be checked that ϕ is an isomorphism. Similar arguments show that $g^*(E)$ and $(h^*(E))|_{B \times \{1\}}$ are isomorphic. Applying the fact gives the result.

Φ is surjective

Gauss Maps

A Gauss map of a k -dimensional vector bundle $p : E \rightarrow B$ in $\mathbb{R}^n$ ($k \leq n \leq \infty$) is a map $g : E \rightarrow \mathbb{R}^n$ such that g is an injective linear map when restricted to fibers.

Example

Let $E_k(\mathbb{R}^n) = \{(V, x) \in G_k(\mathbb{R}^n) \times \mathbb{R}^n \mid x \in V\}$. Then, the projection $q : E_k(\mathbb{R}^n) \rightarrow \mathbb{R}^n$ given by $q(V, x) = x$ is a Gauss map because

$$\begin{aligned} p^{-1}(V) &= \{(V', x) \in G_k(\mathbb{R}^n) \times \mathbb{R}^n \mid x \in V' \text{ and } p(V', x) = V\} \\ &= \{(V', x) \in G_k(\mathbb{R}^n) \times \mathbb{R}^n \mid x \in V' \text{ and } V' = V\} \\ &= \{(V, x) \in G_k(\mathbb{R}^n) \times \mathbb{R}^n \mid x \in V\} \\ &= \{V\} \times V \\ &\cong V \end{aligned}$$

Theorem

For each k -dimensional vector bundle $p : E \rightarrow B$ over a paracompact space B , there exists a Gauss map $g : E \rightarrow \mathbb{R}^\infty$.

This follows from the following fact:

Let $\xi = (E, p, B)$ be a vector bundle over a paracompact space B . Suppose $\{U_\alpha\}$ is an open cover with $\xi|_{U_\alpha}$ trivial. Then there exists a countable open cover $\{W_i\}$ with $\xi|_{W_i}$ trivial.

Proof of Theorem

Let $\{U_i\}$ be a countable open cover as above and let $h_i : p^{-1}(U_i) \rightarrow U_i \times \mathbb{R}^k$ be a local trivialization. Let $\{\psi_i\}$ be a partition of unity subordinate to $\{U_i\}$.

Define $g : E \rightarrow \bigoplus_i \mathbb{R}^k = (\mathbb{R}^k, \mathbb{R}^k, \dots) = \mathbb{R}^\infty$ by $g = (g_1, g_2, \dots)$, where

$$g_i(e) = \begin{cases} (\psi_i \circ p)(e) (pr_2 \circ h_i)(e), & e \in p^{-1}(U_i) \\ 0, & \text{otherwise} \end{cases}$$

Then, g is a Gauss map because each g_i is a Gauss map.

The next result shows that Φ is surjective:

Proposition

Let $\xi = (E, p, B)$ be a k -dimensional vector bundle. There exists a Gauss map $g : E \rightarrow \mathbb{R}^n$ ($k \leq n \leq \infty$) if and only if ξ is isomorphic to $f^*(\gamma_k^n)$ for some map $f : B \rightarrow G_k(\mathbb{R}^n)$.

Φ is injective

The following final result shows that Φ is injective:

Theorem

Let $f_1, f_2 : B \rightarrow G_k(\mathbb{R}^n)$ be two maps such that $f_1^*(\gamma_k^n)$ and $f_2^*(\gamma_k^n)$ are isomorphic. Let $\iota : G_k(\mathbb{R}^n) \rightarrow G_k(\mathbb{R}^{2n})$ be the natural inclusion. Then, $\iota \circ f_1 \simeq \iota \circ f_2$ for $1 \leq n \leq \infty$.

We can now prove a simple classification result.

Classification of Vector Bundles

Two vector bundles over B are isomorphic if and only if their classifying maps are homotopic.

Proof

Let $\xi_1 \cong \xi_2$. Then, there exist $f_i : B \rightarrow G_k(\mathbb{R}^\infty)$, $i = 1, 2$, such that $\xi_i \cong f_i^*(\gamma_k)$.

This gives

$$\Phi([f_1]) = \Phi([f_2]) \implies [f_1] = [f_2],$$

that is, $f_1 \simeq f_2$.

Conversely, if $f_1 \simeq f_2$, then

$$\Phi([f_1]) = \Phi([f_2]) \implies f_1^*(\gamma_k) \cong f_2^*(\gamma_k).$$

However, computing homotopy classes are very difficult.

Sections of a Vector Bundle

Definition

A section of a vector bundle $p : E \rightarrow B$ is a map $s : B \rightarrow E$ such that $p \circ s = \text{id}_B$.

Example

Let M be a smooth manifold. Then sections of TM are vector fields.

Example

Let $p : B \times \mathbb{R}^k \rightarrow B$ be a trivial vector bundle. Then sections are maps $f : B \rightarrow \mathbb{R}^k$.

Example

For any vector bundle $p : E \rightarrow B$, the map $s : B \rightarrow E$ given by $s(b) = 0_b \in p^{-1}(b)$ is a section, called the zero section.

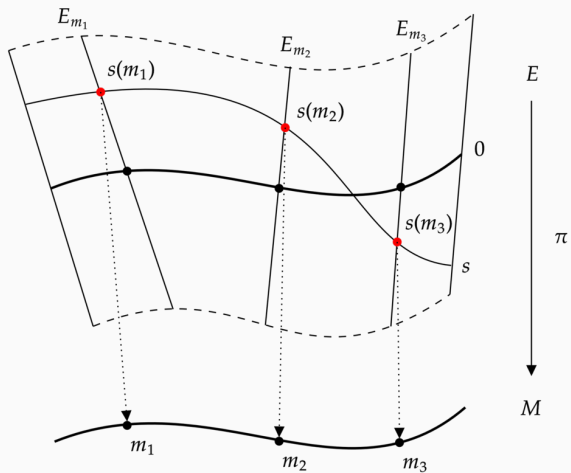


Figure 2: A section of (E, π, M) [4]

Why are Sections Useful?

Theorem

A vector bundle $p : E \rightarrow B$ of rank k is trivial if and only if it admits k sections $s_1, \dots, s_k$ such that $s_1(b), \dots, s_k(b)$ are linearly independent for each $b \in B$.

Proof

Take the standard basis $\{e_1, \dots, e_k\} \subseteq \mathbb{R}^k$ and set $s_i(b) = (b, e_i)$.

Conversely, define $\psi : B \times \mathbb{R}^k \rightarrow E$ by $\psi(b, v_1, \dots, v_k) = v_1 s_1(b) + \dots + v_k s_k(b)$, where $v_i \in \mathbb{R}$.

However, this seems circular.

Defining Characteristic Classes

Recall our Main Theorem: Every rank k bundle ξ is the pullback of the universal bundle γ_k via a classifying map $f : B \rightarrow G_k(\mathbb{R}^\infty)$, i.e.,

$$\xi \cong f^*(\gamma_k).$$

The Construction

It is a known topological fact that the cohomology ring of the infinite Grassmannian is a polynomial algebra:

$$H^*(G_k(\mathbb{R}^\infty); \mathbb{Z}_2) \cong \mathbb{Z}_2[w_1, w_2, \dots, w_k].$$

We define the Stiefel-Whitney classes of ξ to be the pullbacks of these generators:

$$w_i(\xi) := f^*(w_i) \in H^i(B; \mathbb{Z}_2).$$

Properties of Stiefel-Whitney Classes

With this definition, these classes satisfy some key properties:

- (i) Dimension: $w_0(\xi) = 1$ and $w_i(\xi) = 0$ if $i > \text{rank}(\xi)$.
- (ii) Naturality: $w_i(f^*(\xi)) = f^*(w_i(\xi))$ for any map $f : B' \rightarrow B$.
- (iii) Whitney Sum Formula: If ξ and η are bundles over the same base,

$$w(\xi \oplus \eta) = w(\xi) \smile w(\eta)$$

where $w(\xi) = \sum w_i(\xi)$ is the total Stiefel-Whitney class.

Application: Linearly Independent Sections

These axioms provide a powerful algebraic obstruction to the existence of sections.

Theorem

Let ξ be a vector bundle of rank n . If ξ admits k everywhere linearly independent sections, then the top k Stiefel-Whitney classes must vanish:

$$w_{n-k+1}(\xi) = w_{n-k+2}(\xi) = \cdots = w_n(\xi) = 0$$

Proof

If $\xi \cong \epsilon^k \oplus \eta$, where η is a rank $n - k$ vector bundle, then $w(\xi) = w(\epsilon^k \oplus \eta) = w(\epsilon^k) \smile w(\eta) = w(\eta)$. Since $w_i(\eta) = 0$ for $i > n - k$, the result follows.

Summary

1. Vector bundles are complicated and twisted.
2. Instead, we can study homotopy classes of maps into the infinite Grassmannian.
3. That is also hard. Instead, we can study sections of the bundle.
4. That is circular. Instead, we study the cohomology ring of the infinite Grassmannian.

“Algebra is the offer made by the devil to the mathematician. The devil says: I will give you this powerful machine... All you need to do is give me your soul: give up geometry, and you will have this marvelous machine.”

Michael Atiyah

References

- [1] Hatcher, A. (2001), *Algebraic Topology*, available at <https://pi.math.cornell.edu/~hatcher/AT/AT.pdf>
- [2] Hatcher, A. (2017), *Vector Bundles and K-Theory*, available at <https://pi.math.cornell.edu/~hatcher/VBKT/VBpage.html>.
- [3] Husemoller, D. (1993), *Fibre Bundles* (Springer).
- [4] Rudolph G., Schmidt M. (2013). *Differential Geometry and Mathematical Physics Part I. Manifolds, Lie Groups and Hamiltonian Systems* (Springer).
- [5] Wikipedia. *Section (fiber bundle)*, retrieved from [https://en.wikipedia.org/wiki/Section_\(fiber_bundle\)](https://en.wikipedia.org/wiki/Section_(fiber_bundle))
- [6] Wikipedia. *Tangent bundle*, retrieved from https://en.wikipedia.org/wiki/Tangent_bundle.